Design and Analysis of a Two-Stage CMOS OTA Using LTspice

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Abstract

This report presents the design and simulation-based analysis of a two-stage CMOS Operational Transconductance Amplifier (OTA) for a 180nm CMOS process using LTspice. The circuit consists of an NMOS differential input stage with a PMOS active current-mirror load, followed by a PMOS common-source gain stage with an NMOS current sink. Miller compensation, using a compensation capacitor and a series nulling resistor, is used to improve frequency response and stability.

The design process involves small-signal analysis, transistor sizing, bias-current selection, and frequency-compensation calculations. Separate LTspice testbenches are used to evaluate AC response, transient response, common-mode rejection ratio (CMRR), power-supply rejection ratio (PSRR), input common-mode range (ICMR), and output voltage swing. The simulation results are used to assess gain, bandwidth, phase margin, slew rate, settling time, and power consumption against the design requirements.

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Introduction

Operational transconductance amplifiers (OTAs) are fundamental building blocks in analog integrated circuits. They convert an input differential voltage into an output current and are widely used in active filters, sensor interfaces, data converters and other analog signal-processing systems. Their performance is characterized by parameters like open-loop gain, gain-bandwidth product (GBW) , stability, slew rate (SR), input common-mode range (ICMR), and power consumption.

This project focuses on the design and simulation of a two-stage CMOS OTA for a 180nm CMOS process using LTspice. The work involves analytical transistor sizing, small-signal analysis, and frequency compensation, followed by simulations to evaluate the amplifier's performance against the specified design targets.

Design Specifications

The OTA is designed for a single-supply voltage of 1.8 V and a capacitive load of 5 pF, targeting a 180 nm CMOS process. The design specifications define the required gain, frequency response, stability, input common-mode range, transient performance, and power consumption.

Parameter	**Target**	**Description**
Supply voltage (V_{DD})	1.8 V	Single supply
Load capacitance (C_L)	5 pF	Capacitive output load
DC open-loop gain	≥ 60 dB	Low-frequency gain
Gain-bandwidth product	≥ 10 MHz	Unity-gain frequency
Phase margin	$\geq 60°$	Stability criterion
Input common-mode range (ICMR)	0.8–1.6 V	Input operating range
Slew rate	$\geq 5\,\mathrm{V}/\mu\mathrm{s}$	Large-signal response
Settling time	≤ 500 ns	Settling within 0.1%
Power consumption	$\leq 500\,\mu$W	Power Consumed

Table 1: Design specifications of the two-stage CMOS OTA

Topology Description

The designed CMOS OTA consists of two amplification stages, a bias circuit, and a frequency compensation network.

Differential Amplifier

The first stage consists of an NMOS differential input pair with a PMOS active current-mirror load. The differential pair amplifies the voltage difference between the two inputs, while the active load converts the differential current variations into a single-ended output. An NMOS tail-current source establishes the bias current of the differential pair.

Common Source Amplifier

The second stage involves a PMOS common source amplifier with a NMOS current sink. It provides additional voltage gain and drives the capacitive output load.

Frequency Compensation Network

Due to multiple poles in the frequency response of the OTA there is a phase lag, which can reduce stability. So, Miller Compensation is used to improve frequency response as well as phase margin. This technique involves adding a compensation capacitor (C_c) and a resistor (R_z) in series with it. The capacitor splits the poles and pushes the dominant pole towards the origin, i.e. low frequency region and the non-dominant pole is pushed towards high frequency region. However, C_c also gives rise to a RHP (Right Half Plane) Zero, which causes phase lag and reduces phase margin. R_z is used to push this

zero to the high frequency region (ideally ∞) to increase the phase margin and improve the frequency response.

Block Diagram & Circuit Schematic

Below is the block diagram of the Two-Stage CMOS OTA

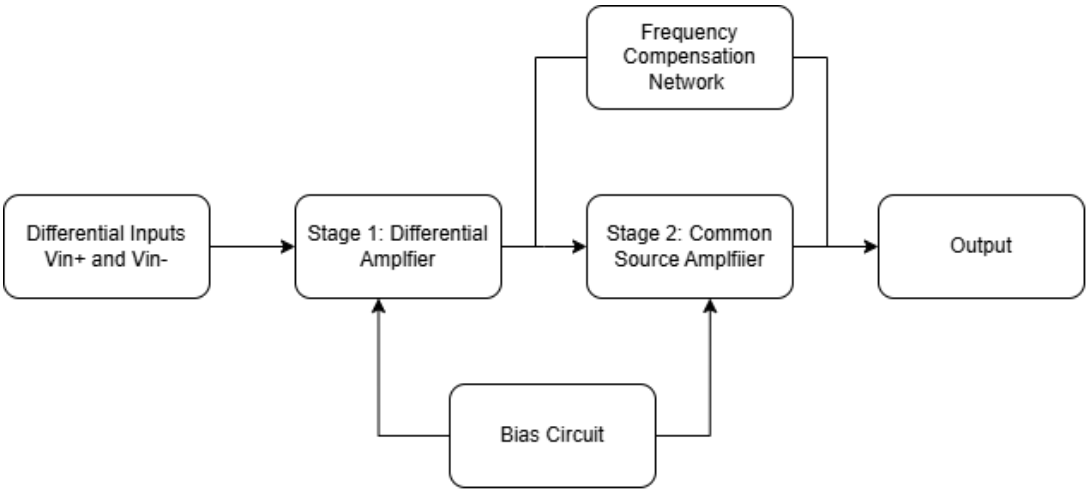


Figure 1: Block diagram of the two-stage CMOS OTA

Below is the complete circuit schematic of the two-stage CMOS OTA, implemented and simulated in LTspice. It shows the differential amplifier, common source amplfiier bias circuit and frequency compensation network. The differential input sources V_{in+} and V_{in-} are applied in simulation testbenches to evaluate the performance of the circuit under different operating conditions.

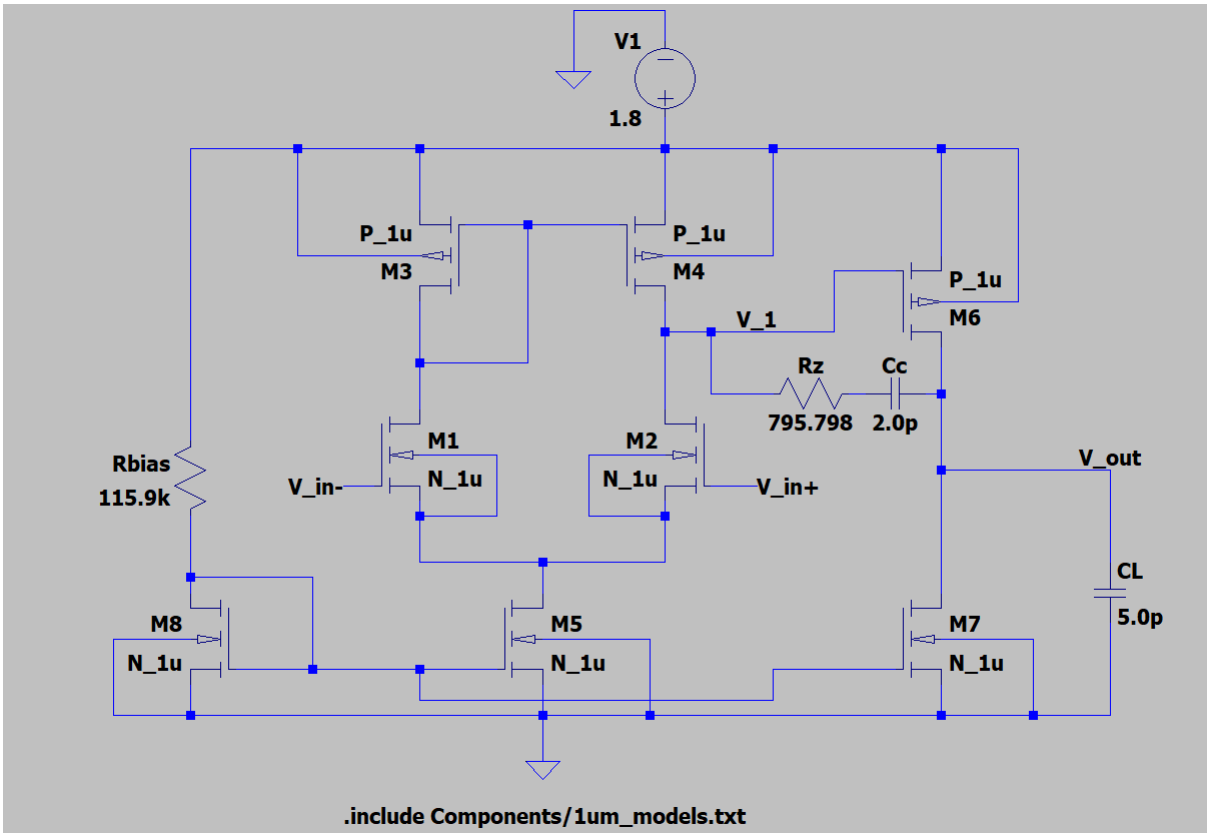


Figure 2: Circuit Schematic

Design Methodology and Analytical Calculations

Design Parameters and Assumptions

The initial transistor sizing is based on the following assumed MOSFET parameters for the target 180 nm CMOS process.

Parameter	Value
NMOS process transconductance parameter ($\mu_n C_{ox}$)	$300\,\mu\text{A/V}^2$
PMOS process transconductance parameter ($\mu_p C_{ox}$)	$60\,\mu\text{A/V}^2$
NMOS threshold voltage ($V_{TH,n}$)	0.4 V
PMOS threshold voltage ($\lvert V_{TH,p}\rvert$)	0.5 V

Table 2: Assumed MOSFET parameters

For the initial transistor sizing, the MOSFET saturation-current equation is used:

$$I_D = \frac{1}{2}\mu C_{ox}\frac{W}{L}V_{OV}^2$$

where

- I_D is the drain current
- W/L is the transistor aspect ratio
- V_{OV} is the overdrive voltage
- μC_{ox} is the process transconductance parameter

The overdrive voltage is defined as follows.
For NMOS transistors,

$$V_{OV,n} = V_{GS,n} - V_{TH,n}$$

For PMOS transistors,

$$V_{OV,p} = V_{SG,p} - |V_{TH,p}|$$

Rearranging the I_D equation allows calculation of $\frac{W}{L}$ using μC_{ox} and V_{OV}

$$\frac{W}{L} = \frac{2I_D}{\mu C_{ox} V_{OV}^2}$$

For PMOS transistors, this becomes

$$\frac{W}{L} = \frac{2I_D}{\mu_p C_{ox} V_{OV,p}^2}$$

The MOSFET transconductance, g_m, is defined as the rate of change of I_D with respect to V_{GS} at a constant V_{DS}:

$$g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right|_{V_{DS}}$$

Using the I_D equation and assuming V_{TH} is constant, g_m can be derived as

$$\begin{aligned} g_m &= \frac{\partial I_D}{\partial V_{GS}} \\ &= \frac{\partial}{\partial V_{GS}}\left(\frac{1}{2}\mu C_{ox}\frac{W}{L}V_{OV}^2\right) \\ &= \mu C_{ox}\frac{W}{L}V_{OV} \end{aligned}$$

Using the I_D equation g_m can be expressed as

$$\boxed{g_m = \frac{2I_D}{V_{OV}}}$$

Alternatively, eliminating V_{OV} gives

$$\boxed{g_m = \sqrt{2\mu C_{ox}\frac{W}{L}I_D}}$$

These equations and process parameters are used in the analytical calculations to determine the component values, transistor aspect ratios, and bias conditions required for the initial circuit design.

Transfer Function Derivation

To derive the transfer function, we need to analyse its small signal equivalent circuit as shown below

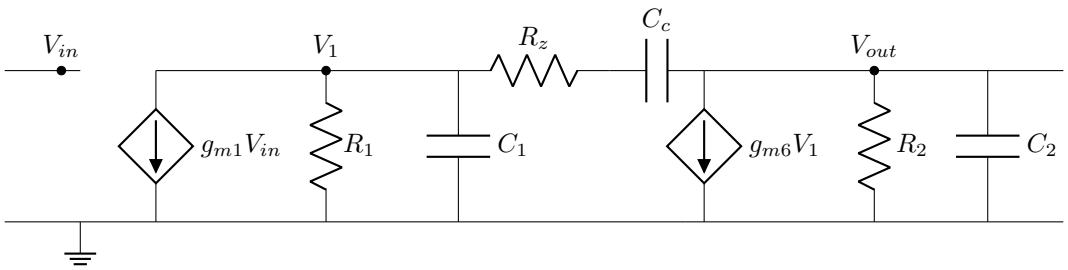


Figure 3: Small-Signal Equivalent Circuit

With reference to the above figure,

- g_{m1} and g_{m6}: Transconductances of the first and second stages, respectively.
- R_1 and R_2: Equivalent output resistances of the two stages.
- C_1 and C_2: Equivalent capacitance at the intermediate node and output node respectively
- C_c and R_z: Miller compensation capacitor and series nulling resistor.
- V_{in}, V_1, and V_{out}: Input, intermediate-node, and output voltages, respectively.

Let $G_1 = 1/R_1$ and $G_2 = 1/R_2$ denote the output conductances of the two stages.
The admittance of the series compensation network is

$$Y_c(s) = \frac{1}{R_z + \frac{1}{sC_c}} = \frac{sC_c}{1 + sR_zC_c}$$

Applying Kirchhoff's Current Law (KCL) at V_1 gives

$$g_{m1}V_{in} + V_1(G_1 + sC_1) + Y_c(V_1 - V_{out}) = 0 \tag{1}$$

Similarly, applying KCL at V_{out} gives

$$g_{m6}V_1 + V_{out}(G_2 + sC_2) + Y_c(V_{out} - V_1) = 0 \tag{2}$$

Rearranging Equation (2),

$$V_1(g_{m6} - Y_c) = -V_{out}(G_2 + sC_2 + Y_c)$$

Therefore,

$$V_1 = -V_{out}\left(\frac{G_2 + sC_2 + Y_c}{g_{m6} - Y_c}\right) \tag{3}$$

Substituting Equation (3) into Equation (1) and simplifying gives:

$$\frac{V_{out}}{V_{in}} = \frac{g_{m1}(g_{m6} - Y_c)}{(G_1 + sC_1)(G_2 + sC_2) + Y_c(G_1 + sC_1 + G_2 + sC_2 + g_{m6})}$$

Substituting the expression for $Y_c(s)$ and multiplying the numerator and denominator by $(1+sR_zC_c)$ results in:

$$\frac{V_{out}}{V_{in}}(s) = \frac{g_{m1}R_1R_2g_{m6}\left[1 - sC_c\left(\frac{1}{g_{m6}} - R_z\right)\right]}{1 + sD_1 + s^2D_2 + s^3D_3}$$

where the denominator coefficients are

$$\begin{aligned} D_1 &= R_1(C_1 + C_c) + R_2(C_2 + C_c) \\ &\quad + g_{m6}R_1R_2C_c + R_zC_c \\ D_2 &= R_1R_2(C_1C_2 + C_1C_c + C_2C_c) \\ &\quad + R_zC_c(R_1C_1 + R_2C_2) \\ D_3 &= R_1R_2R_zC_1C_2C_c \end{aligned}$$

The denominator is third order, giving three poles, while the numerator contains one compensation zero.

DC Gain, Poles, and Compensation Zero

The general three-pole transfer function with a compensation zero is

$$H(s) = \frac{A_{DC}\left(1 - \frac{s}{\omega_z}\right)}{\left(1 + \frac{s}{\omega_{p1}}\right)\left(1 + \frac{s}{\omega_{p2}}\right)\left(1 + \frac{s}{\omega_{p3}}\right)}.$$

Expanding the denominator,

$$\begin{aligned} D(s) = &\left(1 + \frac{s}{\omega_{p1}}\right)\left(1 + \frac{s}{\omega_{p2}}\right)\left(1 + \frac{s}{\omega_{p3}}\right) \\ &= 1 + s\left(\frac{1}{\omega_{p1}} + \frac{1}{\omega_{p2}} + \frac{1}{\omega_{p3}}\right) \\ &\quad + s^2\left(\frac{1}{\omega_{p1}\omega_{p2}} + \frac{1}{\omega_{p1}\omega_{p3}} + \frac{1}{\omega_{p2}\omega_{p3}}\right) \\ &\quad + \frac{s^3}{\omega_{p1}\omega_{p2}\omega_{p3}}. \end{aligned}$$

Comparing this with the circuit characteristic polynomial $1 + sD_1 + s^2D_2 + s^3D_3$, the coefficients are

$$\begin{aligned} D_1 &= \frac{1}{\omega_{p1}} + \frac{1}{\omega_{p2}} + \frac{1}{\omega_{p3}}, \\ D_2 &= \frac{1}{\omega_{p1}\omega_{p2}} + \frac{1}{\omega_{p1}\omega_{p3}} + \frac{1}{\omega_{p2}\omega_{p3}}, \\ D_3 &= \frac{1}{\omega_{p1}\omega_{p2}\omega_{p3}}. \end{aligned}$$

Since $\omega_{p1} \ll \omega_{p2} \ll \omega_{p3}$, ω_{p1} is the low-frequency dominant pole. Therefore,

$$D_1 \approx \frac{1}{\omega_{p1}} \quad \Rightarrow \quad \omega_{p1} \approx \frac{1}{D_1}.$$

For the second pole, the dominant term in D_2 is $1/(\omega_{p1}\omega_{p2})$. Hence,

$$D_2 \approx \frac{1}{\omega_{p1}\omega_{p2}} \quad \Rightarrow \quad \omega_{p2} \approx \frac{1}{\omega_{p1}D_2}.$$

For the third pole,

$$D_3 = \frac{1}{\omega_{p1}\omega_{p2}\omega_{p3}},$$

which gives

$$\omega_{p3} \approx \frac{1}{\omega_{p1}\omega_{p2}D_3} \approx \frac{D_2}{D_3}.$$

The DC open-loop gain is obtained by substituting $s = 0$ in the transfer function:

$$A_{v0} = g_{m1}R_1 g_{m6}R_2.$$

The first coefficient is

$$D_1 = R_1(C_1 + C_c) + R_2(C_2 + C_c) + g_{m6}R_1R_2C_c + R_zC_c.$$

As R_1R_2 is very large compared to other terms, $g_{m6}R_1R_2C_c$ dominates,

$$D_1 \approx g_{m6}R_1R_2C_c.$$

Therefore, the dominant pole is

$$\boxed{\omega_{p1} \approx \frac{1}{g_{m6}R_1R_2C_c}}.$$

The second coefficient is

$$\begin{aligned} D_2 = {} & R_1R_2(C_1C_2 + C_1C_c + C_2C_c) \\ & + R_zC_c(R_1C_1 + R_2C_2). \end{aligned}$$

As R_1R_2 is very large,

$$D_2 = R_1R_2(C_1C_2 + C_1C_c + C_2C_c)$$

As $C_1 \ll C_c, C_2$,

$$D_2 \approx R_1R_2C_2C_c.$$

Substituting the approximation for ω_{p1} gives

$$\omega_{p2} \approx \frac{g_{m6}}{C_2}.$$

Since $C_2 \approx C_L$,

$$\boxed{\omega_{p2} \approx \frac{g_{m6}}{C_L}}.$$

The third coefficient is

$$D_3 = R_1R_2R_zC_1C_2C_c.$$

Using $\omega_{p3} \approx D_2/D_3$ and $D_2 \approx R_1R_2C_2C_c$,

$$\boxed{\omega_{p3} \approx \frac{1}{R_zC_1}}.$$

Since C_1 is small, the third pole is expected to occur at a high frequency.
The compensation zero is obtained by setting the numerator equal to zero:

$$\omega_z = \frac{1}{C_c\left(\frac{1}{g_{m6}} - R_z\right)}.$$

Based on the value of R_z

- For $R_z < 1/g_{m6}$, the zero lies in the right-half plane (RHP),
- for $R_z > 1/g_{m6}$, zero lies in the left-half plane (LHP).
- At $R_z = 1/g_{m6}$, the zero moves to ∞

Analytical Design Calculations

Although the minimum length that can be manufactured in a 180 nm process is $L_{min} = 0.18\,\mu$m, for the initial design, a channel length of $L = 1\,\mu$m is assumed for all transistors. This is done deliberately to decrease short channel effects and increase r_o for each transistor.

The transistor widths are calculated from their respective aspect ratios using

$$W = (\frac{W}{L})L$$

Compensation Capacitance Selection

As a design guideline, the compensation capacitance is selected based on:

$$C_c \geq 0.22C_L$$

For the specified load capacitance $C_L = 5\,\text{pF}$,

$$C_c \geq 0.22(5\,\text{pF}) = 1.1\,\text{pF}.$$

So, the compensation capacitor is selected as

$$\boxed{C_c = 2\,\text{pF}}.$$

Slew-Rate-Based Tail Current Calculation

The slew rate (SR) is the maximum rate of change of the output voltage with respect to time:

$$SR = \left|\frac{dV_{out}}{dt}\right|_{\max}.$$

The current through a capacitor is given by

$$i_C = C\frac{dV_C}{dt}.$$

In a Miller-compensated OTA, the slew rate is limited by the current available to charge or discharge the compensation capacitor C_c. Assuming the first-stage tail current limits this current,

$$I_{tail} \approx C_c SR.$$

Hence, the minimum tail current required for the specified slew rate is

$$I_{tail,\min} = SR_{\min}C_c.$$

For $SR_{\min} = 5\,\text{V}/\mu\text{s}$ and $C_c = 2\,\text{pF}$,

$$\begin{aligned} I_{tail,\min} &= (5 \times 10^6)(2 \times 10^{-12}) \\ &= 10\,\mu\text{A}. \end{aligned}$$

A tail current twice the calculated minimum is selected to provide a nominal design margin:

$$\boxed{I_{tail} = 20\,\mu\text{A}}.$$

Differential Pair Sizing

At zero differential input, the tail current divides equally between the matched NMOS input transistors. Hence,

$$I_{D1} = I_{D2} = \frac{I_{\text{tail}}}{2} = \frac{20\,\mu\text{A}}{2} = 10\,\mu\text{A}.$$

For a Miller-compensated two-stage OTA, the gain-bandwidth product is approximated by

$$GBW \approx \frac{g_{m1}}{2\pi C_c}.$$

For the target $GBW = 10\,\text{MHz}$ and $C_c = 2\,\text{pF}$,

$$\begin{aligned} g_{m1} &= 2\pi(GBW)C_c \\ &= 2\pi(10 \times 10^6)(2 \times 10^{-12}) \\ &= 125.66\,\mu\text{S}. \end{aligned}$$

Using the MOSFET saturation-current relationship, the aspect ratio of each input transistor is

$$\begin{aligned} \left(\frac{W}{L}\right)_{1,2} &= \frac{g_{m1}^2}{2\mu_n C_{ox} I_{D1}} \\ &= \frac{(125.66 \times 10^{-6})^2}{2(300 \times 10^{-6})(10 \times 10^{-6})} \\ &\approx 2.63. \end{aligned}$$

As $L = 1\,\mu\text{m}$,

$$\boxed{W_1 = W_2 = 2.63\,\mu\text{m}}.$$

PMOS Active Load Sizing

The PMOS transistors M_3 and M_4 form a current-mirror load. At the balanced DC operating point, assuming matched transistors and an ideal current mirror,

$$I_{D3} = I_{D4} = 10\,\mu\text{A}.$$

Considering a PMOS overdrive voltage of $V_{OV,p} = 0.15\,\text{V}$ is selected. Using the I_D equation, the required aspect ratio is

$$\begin{aligned} \left(\frac{W}{L}\right)_{3,4} &= \frac{2I_D}{\mu_p C_{ox} V_{OV,p}^2} \\ &= \frac{2(10 \times 10^{-6})}{(60 \times 10^{-6})(0.15)^2} \\ &\approx 14.81. \end{aligned}$$

Since $L = 1\,\mu\text{m}$, the transistor widths are

$$\boxed{W_3 = W_4 = 14.81\,\mu\text{m}}.$$

Tail Current Transistor Sizing

The NMOS transistor M_5 provides the tail current for the differential input pair. The overdrive voltage of the input transistor, V_{OV1}, is first calculated using

$$V_{OV1} = \frac{2I_{D1}}{g_{m1}}.$$

Using $I_{D1} = 10\,\mu\text{A}$ and $g_{m1} = 125.66\,\mu\text{S}$,

$$\begin{aligned} V_{OV1} &= \frac{2(10 \times 10^{-6})}{125.66 \times 10^{-6}} \\ &\approx 0.159\,\text{V}. \end{aligned}$$

For M_5, the drain voltage is approximately

$$V_{D5} = V_{in} - V_{GS1}.$$

For M_5 to remain in saturation, its drain-source voltage must satisfy

$$V_{DS5} \geq V_{OV5}.$$

Since the source of M_5 is connected to ground, $V_{DS5} = V_{D5}$. Therefore,

$$V_{in} - V_{GS1} \geq V_{OV5}.$$

Rearranging,

$$V_{in} \geq V_{GS1} + V_{OV5}.$$

For the input transistor M_1,

$$V_{GS1} = V_{TH,n} + V_{OV1}.$$

Substituting this into the saturation condition gives

$$V_{in} \geq V_{TH,n} + V_{OV1} + V_{OV5}.$$

At the minimum input common-mode voltage, $V_{in} = ICMR^-$, the condition becomes

$$\boxed{ICMR^- = V_{TH,n} + V_{OV1} + V_{OV5}}.$$

For the specified $ICMR^-$ of 0.8 V, with $V_{TH,n} = 0.4\,\text{V}$,

$$\begin{aligned} V_{OV5} &= ICMR^- - V_{TH,n} - V_{OV1} \\ &= 0.8 - 0.4 - 0.159 \\ &= 0.241\,\text{V}. \end{aligned}$$

Since M_5 carries the total tail current, $I_{D5} = I_{\text{tail}} = 20\,\mu\text{A}$. Its aspect ratio is calculated using the saturation-current equation:

$$\begin{aligned} \left(\frac{W}{L}\right)_5 &= \frac{2I_{D5}}{\mu_n C_{ox} V_{OV5}^2} \\ &= \frac{2(20 \times 10^{-6})}{(300 \times 10^{-6})(0.241)^2} \\ &\approx 2.30. \end{aligned}$$

Since $L = 1\,\mu\text{m}$, the transistor width is

$$\boxed{W_5 = 2.30\,\mu\text{m}}.$$

Second-Stage Transistor Sizing

For the balanced DC operating point, the differential input voltage is zero, and the two input transistors carry equal currents. Assuming ideal operation of the PMOS current-mirror load, the drain voltages of M_3 and M_4 are taken to be equal:

$$V_{D3} = V_{D4}.$$

Since M_3 is diode-connected and M_3 and M_4 share the same gate connection,

$$V_{G3} = V_{D3}, \qquad V_{G3} = V_{G4}.$$

Therefore,

$$V_{G4} = V_{D3} = V_{D4}.$$

The gate of M_6 is connected to the drain of M_4, giving

$$V_{G6} = V_{D4}.$$

Hence,

$$\boxed{V_{G4} = V_{G6}}.$$

Both M_4 and M_6 have their sources connected to $V_{DD} = 1.8\,\text{V}$, so

$$V_{S4} = V_{S6} = 1.8\,\mathrm{V}.$$

Consequently, their V_{SG} are equal. Assuming identical $V_{TH,p}$,

$$\boxed{V_{OV4} = V_{OV6}},$$

where $V_{OV,p} = V_{SG,p} - |V_{TH,p}|$.
These relationships are used as assumptions for the initial analytical sizing.
The g_m of a MOSFET operating in saturation under the square-law model is

$$g_m = \sqrt{2\mu C_{ox}\frac{W}{L}I_D}.$$

For M_4, the g_m is

$$g_{m4} = \sqrt{2\mu_p C_{ox}\left(\frac{W}{L}\right)_4 I_{D4}}.$$

Using $\mu_p C_{ox} = 60\,\mu\mathrm{A/V}^2$, $(W/L)_4 = 14.81$, and $I_{D4} = 10\,\mu\mathrm{A}$,

$$\begin{aligned} g_{m4} &= \sqrt{2(60\times10^{-6})(14.81)(10\times10^{-6})} \\ &\approx 133.31\,\mu\mathrm{S}. \end{aligned}$$

To obtain a phase margin greater than approximately 60°, a commonly used initial design guideline is to choose the g_m of the second-stage transistor as ten times that of the input transistor in the differential pair:

$$g_{m6} = 10 g_{m1}.$$

Using $g_{m1} = 125.66\,\mu\mathrm{S}$,

$$\begin{aligned} g_{m6} &= 10(125.66\,\mu\mathrm{S}) \\ &= 1.2566\,\mathrm{mS}. \end{aligned}$$

Applying the previously established g_m relationship to M_4 and M_6,

$$g_{m4} = \frac{2I_{D4}}{V_{OV4}}, \qquad g_{m6} = \frac{2I_{D6}}{V_{OV6}}.$$

As $V_{OV4} = V_{OV6}$,

$$\frac{g_{m6}}{g_{m4}} = \frac{I_{D6}}{I_{D4}}.$$

From the I_D equation,

$$I_D = \frac{1}{2}\mu_p C_{ox}\frac{W}{L}V_{OV}^2.$$

Because M_4 and M_6 have the same $\mu_p C_{ox}$ and V_{OV},

$$\frac{I_{D6}}{I_{D4}} = \frac{(W/L)_6}{(W/L)_4}.$$

Combining the two ratios gives

$$\frac{g_{m6}}{g_{m4}} = \frac{(W/L)_6}{(W/L)_4}.$$

Therefore, the aspect ratio of M_6 is

$$\boxed{\left(\frac{W}{L}\right)_6 = \frac{g_{m6}}{g_{m4}}\left(\frac{W}{L}\right)_4}.$$

Substituting the calculated values,

$$\left(\frac{W}{L}\right)_6 = \frac{1.2566\,\text{mS}}{133.31\,\mu\text{S}} \quad (14.81)$$
$$\approx 139.59.$$

Therefore, considering $L = 1\,\mu\text{m}$,

$$\boxed{W_6 \approx 139.59\,\mu\text{m}}.$$

Second-Stage Current Sink Sizing

The NMOS transistor M_7 provides the current sink for the second stage. Its drain current is selected to match the drain current of M_6:

$$I_{D7} = I_{D6}.$$

The drain current of M_6 is calculated using its g_m and aspect ratio:

$$I_{D6} = \frac{g_{m6}^2}{2\mu_p C_{ox}(W/L)_6}.$$

Using $g_{m6} = 1.2566\,\text{mS}$, $\mu_p C_{ox} = 60\,\mu\text{A/V}^2$, and $(W/L)_6 = 139.59$,

$$I_{D6} = \frac{(1.2566 \times 10^{-3})^2}{2(60 \times 10^{-6})(139.59)}$$
$$\approx 94.3\,\mu\text{A}.$$

Therefore,

$$\boxed{I_{D7} = I_{D6} \approx 94.3\,\mu\text{A}}.$$

Since M_7 and the tail transistor M_5 share the same V_G and V_S, their V_{OV} are assumed equal:

$$V_{OV7} = V_{OV5} = 0.241\,\text{V}.$$

Using the saturation-current equation for both transistors, their current ratio is

$$\frac{I_{D7}}{I_{D5}} = \frac{(W/L)_7}{(W/L)_5}.$$

Rearranging,

$$\left(\frac{W}{L}\right)_7 = \left(\frac{W}{L}\right)_5 \frac{I_{D7}}{I_{D5}}.$$

Substituting $(W/L)_5 = 2.30$, $I_{D7} = 94.3\,\mu\text{A}$, and $I_{D5} = 20\,\mu\text{A}$,

$$\left(\frac{W}{L}\right)_7 = 2.30\left(\frac{94.3}{20}\right)$$
$$\approx 10.84.$$

As $L = 1\,\mu\text{m}$,

$$\boxed{W_7 \approx 10.84\,\mu\text{m}}.$$

Bias Reference Transistor Sizing

The NMOS transistor M_8 generates the reference bias voltage used to establish the tail current through M_5. As that M_8 and M_5 share the same V_G and V_S, their V_{OV} are assumed equal:

$$V_{OV8} = V_{OV5} = 0.241\,\text{V}.$$

A reference current of $I_{\text{ref}} = 10\,\mu\text{A}$ is selected for M_8. Since M_8 and M_5 have same V_{OV}, their current ratio is given by their aspect ratios:

$$\frac{I_{\text{ref}}}{I_{D5}} = \frac{(W/L)_8}{(W/L)_5}.$$

Therefore,

$$\left(\frac{W}{L}\right)_8 = \left(\frac{W}{L}\right)_5 \frac{I_{\text{ref}}}{I_{D5}}.$$

Using $(W/L)_5 = 2.30$, $I_{\text{ref}} = 10\,\mu\text{A}$, and $I_{D5} = 20\,\mu\text{A}$,

$$\begin{aligned}\left(\frac{W}{L}\right)_8 &= 2.30\left(\frac{10}{20}\right)\\ &= 1.15.\end{aligned}$$

Since $L = 1\,\mu\text{m}$, the width of M_8 is

$$\boxed{W_8 = 1.15\,\mu\text{m}}.$$

The V_{GS} of M_8 is calculated as

$$\begin{aligned}V_{GS8} &= V_{TH,n} + V_{OV8}\\ &= 0.4 + 0.241\\ &= 0.641\,\text{V}.\end{aligned}$$

As $V_{S8} = 0$,

$$V_{G8} = 0.641\,\text{V}.$$

As M_8 is diode-connected,

$$\begin{aligned}V_{D8} &= V_{G8}\\ &= 0.641\,\text{V}.\end{aligned}$$

R_{bias} can be calculated by:

$$R_{\text{bias}} = \frac{V_{DD} - V_{D8}}{I_{\text{ref}}}.$$

Substituting $V_{DD} = 1.8\,\text{V}$, $V_{D8} = 0.641\,\text{V}$, and $I_{\text{ref}} = 10\,\mu\text{A}$,

$$\begin{aligned}R_{\text{bias}} &= \frac{1.8 - 0.641}{10 \times 10^{-6}}\\ &\approx 115.9\,\text{k}\Omega.\end{aligned}$$

Hence, the initial bias circuit parameters are

$$\boxed{(W_8 = 1.15\,\mu\text{m}, \qquad R_{\text{bias}} \approx 115.9\,\text{k}\Omega}.$$

Nulling Resistor

For the compensation zero to lie at ∞, the nulling resistor is selected as

$$R_z = \frac{1}{g_{m6}}.$$

Substituting $g_{m6} = 1.2566\,\text{mS}$,

$$R_z = \frac{1}{1.2566 \times 10^{-3}} \approx 795.798\,\Omega.$$

Therefore, the nulling resistor is

$$\boxed{R_z = 795.798\,\Omega}.$$

Summary of Design Parameters

The final transistor dimensions and passive component values obtained from the analytical design calculations are summarized below.

Transistor	**Function**	**Type**	**Width (W) (μm)**	**Length (L) (μm)**
M1, M2	Differential input pair	NMOS	2.63	1.00
M3, M4	Active current-mirror load	PMOS	14.81	1.00
M5	Tail current source	NMOS	2.30	1.00
M6	Second-stage amplifier	PMOS	139.60	1.00
M7	Second-stage current sink	NMOS	10.84	1.00
M8	Bias transistor	NMOS	1.15	1.00

Table 3: Final transistor dimensions used in the OTA design

Below are the passive components values used in the design

Component	**Value**	**Function**
R_{bias}	$115.9\,\text{k}\Omega$	Bias generation
R_z	$795.8\,\Omega$	Frequency compensation
C_c	$2.0\,\text{pF}$	Miller compensation

Table 4: Passive Component Values

Simulation Results and Analysis

Simulation Methodology

The designed two-stage CMOS OTA evaluated using LTspice to determine its performance characteristics.

As all MOSFETs use a channel length of $L = 1\,\mu$m, a file containing models `1um_models.txt` was defined using the assumed design parameters which are mentioned earlier.

Separate testbenches were developed to analyze the DC operating point, AC response, transient response, input common-mode range (ICMR), output voltage swing, common-mode rejection ratio (CMRR), and power-supply rejection ratio (PSRR). Each testbench was configured for its respective analysis using appropriate input sources, circuit connections, and simulation directives.

DC Operating Point Analysis

A DC operating-point analysis was performed using `testbench_operating_point.asc`, with both input terminals biased at 0.9 V. This analysis was used to evaluate the transistor bias currents and quiescent power consumption.

The schematic of the testbench used for the analysis is shown below:

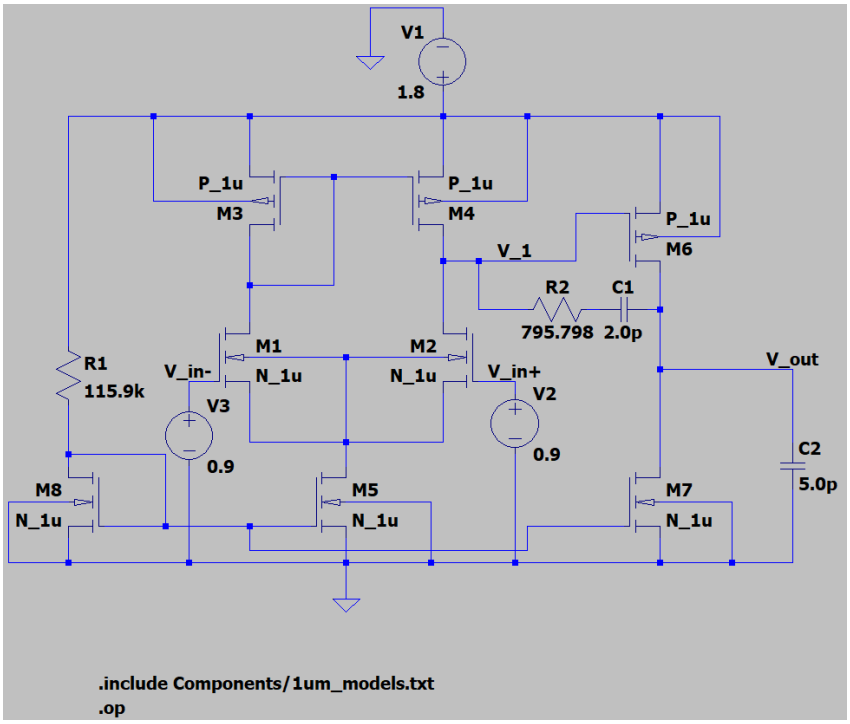


Figure 4: DC Operating Point Testbench

The table below compares the analytically calculated parameters with the values obtained from LTspice.

Parameter	**Calculated**	**Simulated**
Input pair current ($I_{D1} = I_{D2}$)	$10.00\,\mu$A	$9.789\,\mu$A
Tail current (I_{D5})	$20.00\,\mu$A	$19.578\,\mu$A
Second-stage current ($I_{D6} = I_{D7}$)	$94.30\,\mu$A	$96.952\,\mu$A
Bias reference current (I_{D8})	$10.00\,\mu$A	$10.059\,\mu$A
Quiescent supply current (I_{Supply})	$\leq 277.78\,\mu$A	$126.589\,\mu$A
DC power consumption (P_{DC})	$\leq 500\,\mu$W	$227.86\,\mu$W

Table 5: Calculated and simulated DC Operating Point Results

The simulated input-pair currents are approximately equal, as expected for equal input voltages. The tail current is also approximately equal to the sum of the two input branch currents.

The differences between calculated and simulated currents are due to the non-ideal device characteristics represented by the LTspice models.

The maximum allowable supply current is obtained from the specified power limit as

$$\begin{aligned} I_{\text{supply,max}} &= \frac{P_{\text{DC,max}}}{V_{DD}} \\ &= \frac{500\,\mu\text{W}}{1.8\,\text{V}} \\ &= 277.78\,\mu\text{A} \end{aligned}$$

The DC power consumption is calculated using the simulated supply current as

$$\begin{aligned} P_{\text{DC}} &= V_{DD} I_{\text{supply}} \\ &= 1.8\,\text{V} \times 126.589\,\mu\text{A} \\ &= 227.86\,\mu\text{W} \end{aligned}$$

AC Analysis

An AC analysis was performed using `testbench_ac.asc` to evaluate the frequency response and stability of the designed OTA. The DC gain, gain-bandwidth product (GBW), and phase margin (PM) were obtained from the simulated frequency response.

The testbench schematic used for the analysis is shown in Fig.5.

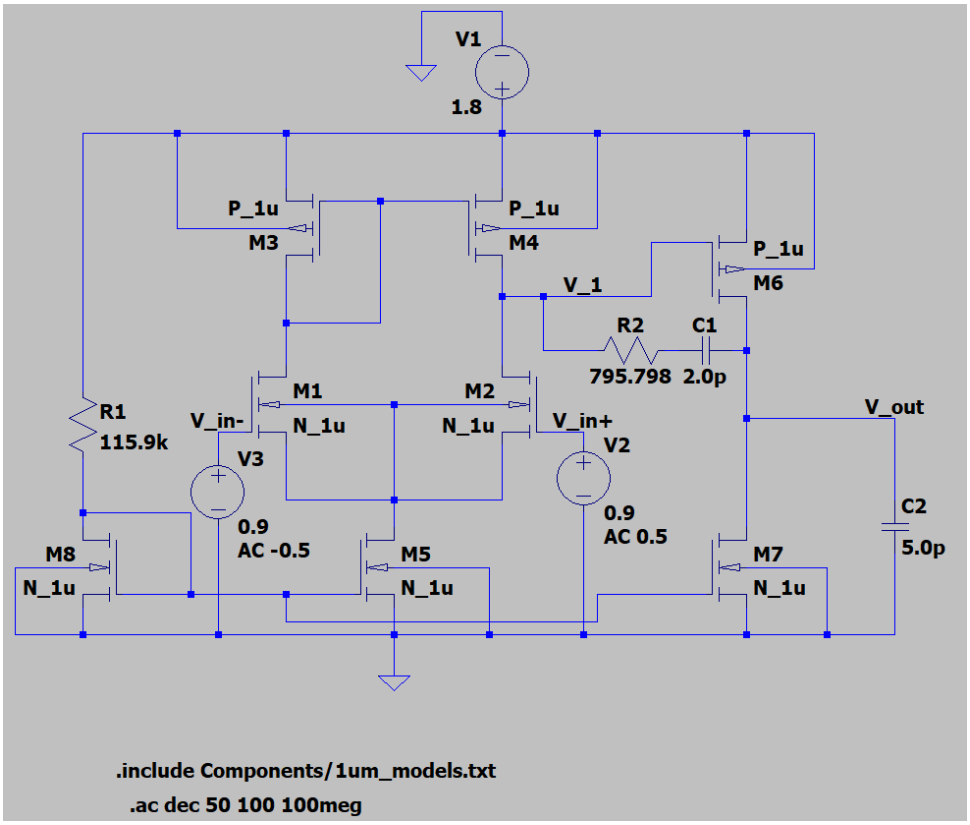


Figure 5: AC Analysis Testbench

The simulated AC frequency response is shown in Fig.6. The first cursor, placed at 100 Hz, gives the low-frequency gain, which is approximately equal to the DC gain. The second cursor is placed near the unity-gain crossover to determine the GBW and phase margin.

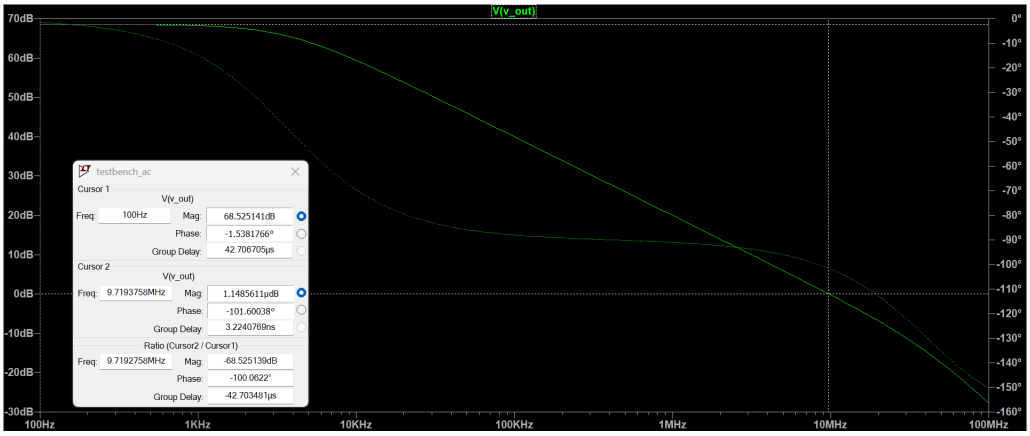


Figure 6: Simulated AC Frequency Response

The DC gain obtained at 100 Hz is 68.525 dB. The GBW is obtained from the frequency at which the magnitude response crosses approximately 0 dB. From the second cursor, the crossover frequency is 9.719 MHz, with a phase of $-101.6°$.

The GBW is therefore

$$\begin{aligned} \text{GBW} &\approx f_u \\ &= 9.719\,\text{MHz} \end{aligned}$$

The phase margin is calculated at the unity-gain crossover frequency as

$$\begin{aligned} PM &= 180^\circ + |\angle A(j\omega_u)| \\ &= 180^\circ - 101.6^\circ \\ &= 78.4^\circ \end{aligned}$$

The simulated results are summarized in Table 6.

Parameter	**Target**	**Simulated**
DC gain	$\geq 60\,\text{dB}$	$68.525\,\text{dB}$
GBW	$\geq 10\,\text{MHz}$	$9.719\,\text{MHz}$
Phase margin	$\geq 60^\circ$	78.4°

Table 6: AC Performance

The simulated DC gain and phase margin satisfy their respective targets. The GBW is slightly below the specified target of 10 MHz, while the phase margin of 78.4° indicates good phase stability for the specified capacitive load.

Transient Response Analysis

A transient analysis was performed using `testbench_transient.asc` to evaluate the large-signal response of the designed OTA. The OTA was configured as a unity-gain buffer by connecting $V_{\text{in}-}$ to V_{out}. A square wave switching between 0.65 V and 1.15 V with a 2 μs pulse width and 1 ns rise/fall times was applied to $V_{\text{in}+}$ to evaluate the step response.

The schematic for this is illustrated in Fig. 7.

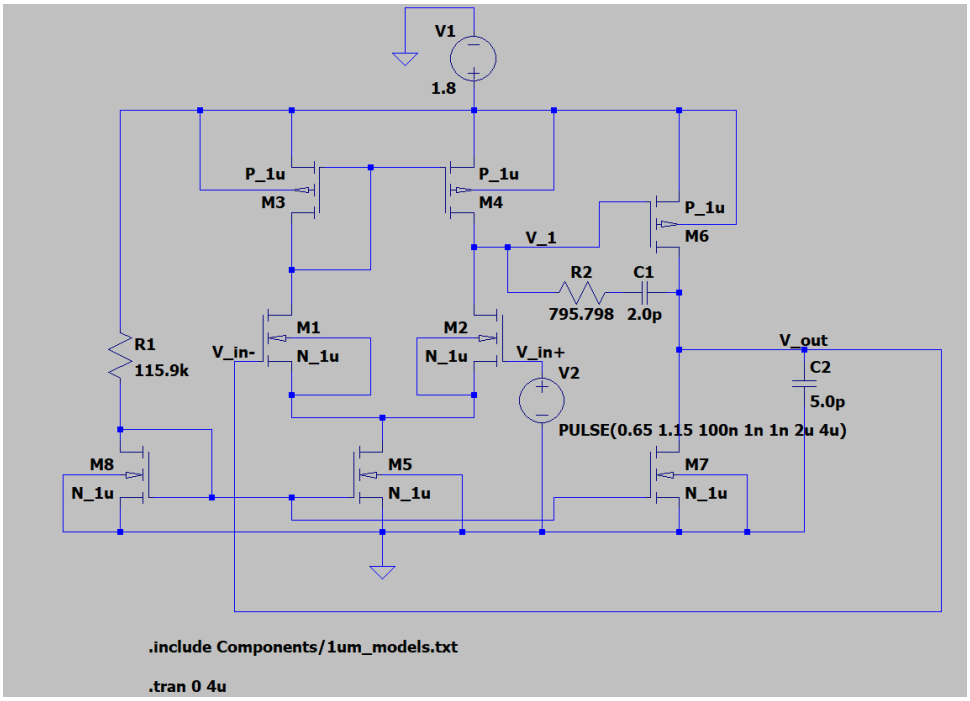


Figure 7: Transient Response Analysis Testbench

The simulated transient response waveforms are shown in Fig. 8.

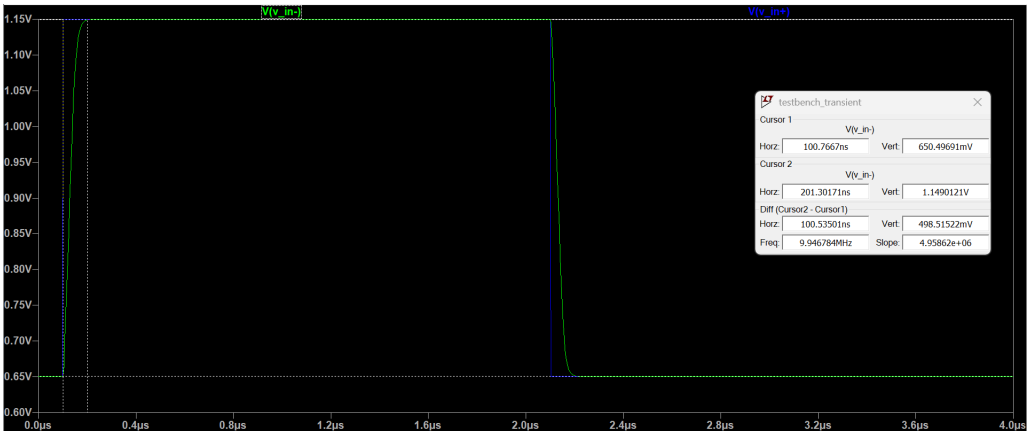


Figure 8: Simulated Transient Response

The slew rate was calculated from the slope of the output transition using the LTspice waveform cursors. By placing the cursors at 100.77 ns and 201.30 ns, a transition slope of 4.959×10^6 V/s was extracted:

$$\begin{aligned} SR &= \left| \frac{\Delta V_{\text{out}}}{\Delta t} \right| \\ &= 4.959 \times 10^6 \, \text{V/s} \\ &= 4.959 \, \text{V}/\mu\text{s} \end{aligned}$$

The settling time was measured as the time required for the output voltage to enter and remain within a $\pm 0.1\%$ error band of its final steady-state value following the input transition. From the cursor difference data, the settling time was determined to be 100.53 ns.

The transient performance metrics are summarized and compared against design specifications in Table 7.

Parameter	**Target**	**Simulated**
Slew Rate	$\geq 5 \, \text{V}/\mu\text{s}$	$4.959 \, \text{V}/\mu\text{s}$
Settling Time (0.1%)	≤ 500 ns	100.53 ns

Table 7: Transient Performance

While the simulated settling time easily satisfies the design specification of 500 ns, the measured slew rate falls slightly short of the $5 \, \text{V}/\mu\text{s}$ target by approximately 0.82%.

Input Common-Mode Range (ICMR) and Output Voltage Swing Analysis

The Input Common-Mode Range (ICMR) and Output Voltage Swing are defined as:

- The Input Common-Mode Range (ICMR) is the range of input voltage over which the OTA functions properly.
- The Output Voltage Swing is the range of output voltage over which the OTA can cleanly deliver a signal without clipping.

The Input Common-Mode Range (ICMR) and output voltage swing of the designed OTA were evaluated by observing its DC transfer characteristics using `testbench_icmr_and_output_swing.asc`.

The OTA was configured as a unity-gain buffer by feeding back the output to the inverting input ($V_{\text{in}-}$). A DC sweep was applied to $V_{\text{in}+}$ from 0 V to 1.8 V in steps of 0.01 V to determine the input range over which the output tracks the input with an error of approximately 1% or less.

The schematic of the testbench used for this analysis is shown in Fig. 9.

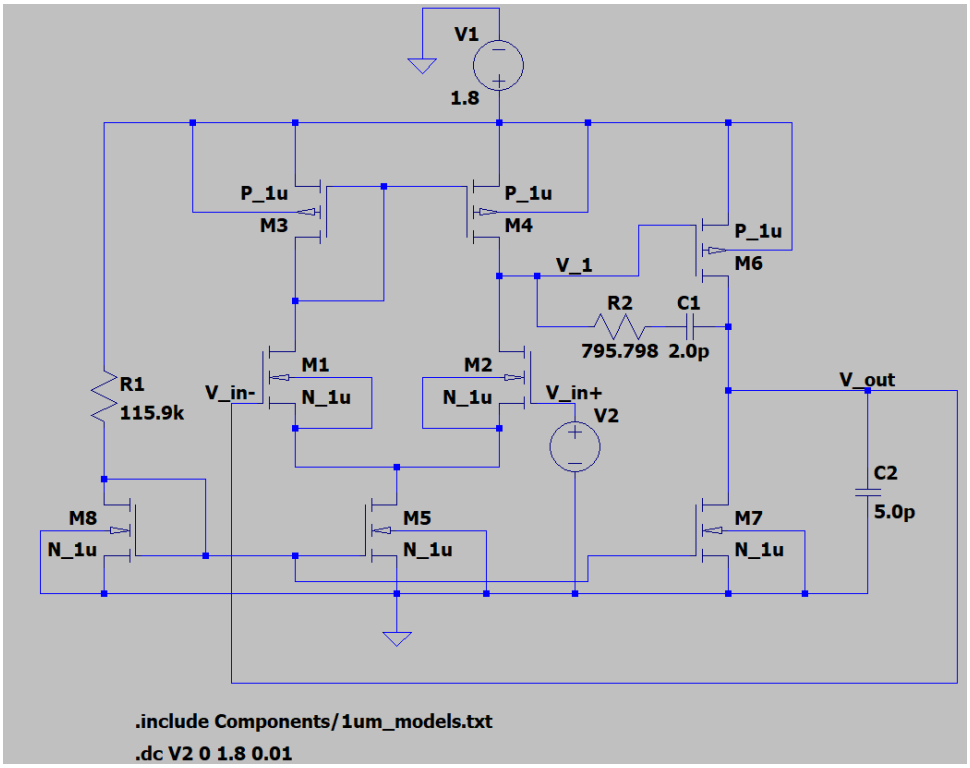


Figure 9: ICMR and Output Swing Testbench

The simulated DC transfer characteristics are shown in Fig. 10.

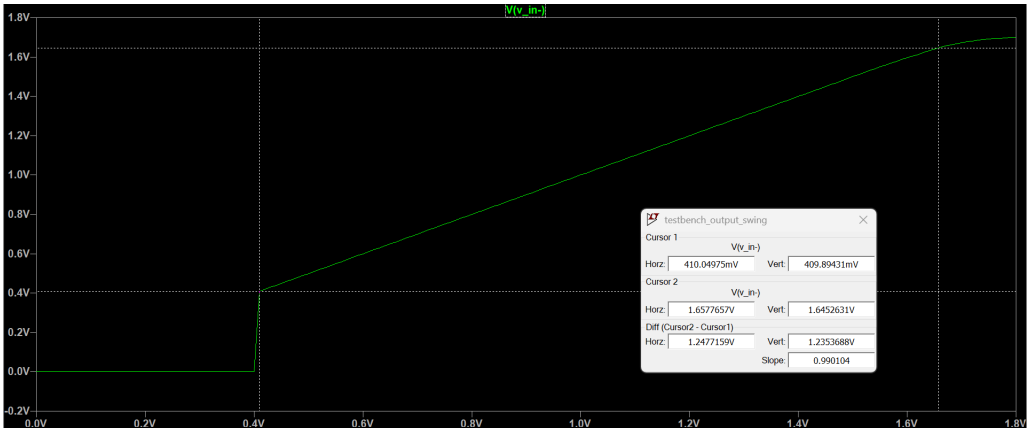


Figure 10: DC Transfer Characteristics for ICMR and Output Swing

As shown in Fig. 10, the horizontal axis represents the input voltage ($V_{\text{in+}}$), which determines the input common-mode range (ICMR), while the vertical axis represents the output voltage (V_{out}).

From the simulated DC transfer characteristics, the ICMR is approximately 0.410–1.658 V, and the output voltage swing is approximately 0.410–1.645 V.

The results are summarized in Table 8.

Parameter	**Target**	**Simulated**
Input common-mode range	0.8–1.6 V	0.410–1.658 V
Output voltage swing	–	0.410–1.645 V

Table 8: ICMR and Output Swing Results

The simulated ICMR covers the specified input common-mode range of 0.8–1.6 V. The output voltage swing was measured to be approximately 0.410–1.645 V under the simulated conditions.

Common-Mode Rejection Ratio (CMRR) Analysis

The Common-Mode Rejection Ratio (CMRR) measures the ability of an OTA to amplify the differential component of its input while rejecting signals common to both input terminals. A higher CMRR indicates better rejection of common-mode signals, such as noise appearing equally at both inputs.

The CMRR is defined as the ratio of the differential-mode gain (A_d) to the common-mode differential gain (A_{cmdm}):

$$\mathrm{CMRR} = \left| \frac{A_d}{A_{cmdm}} \right|$$

Expressing the ratio in decibels gives

$$\begin{aligned} \mathrm{CMRR}_{\mathrm{dB}} &= 20 \log_{10} \left| \frac{A_d}{A_{cmdm}} \right| \\ &= 20 \log_{10} |A_d| - 20 \log_{10} |A_{cmdm}| \\ &= A_{d,\mathrm{dB}} - A_{cmdm,\mathrm{dB}} \end{aligned}$$

So, $CMRR_{dB}$ can be calculated using $A_{d,dB}$ and $A_{cmdm,dB}$ provided both gains are measured at the same frequency.

The CMRR was evaluated using `testbench_cmrr.asc`. For the common-mode gain measurement, $V_{\mathrm{in}+}$ and $V_{\mathrm{in}-}$ were connected together and biased at 0.9 V, with a small-signal AC input of 1 V. The testbench schematic is shown in Fig. 11.

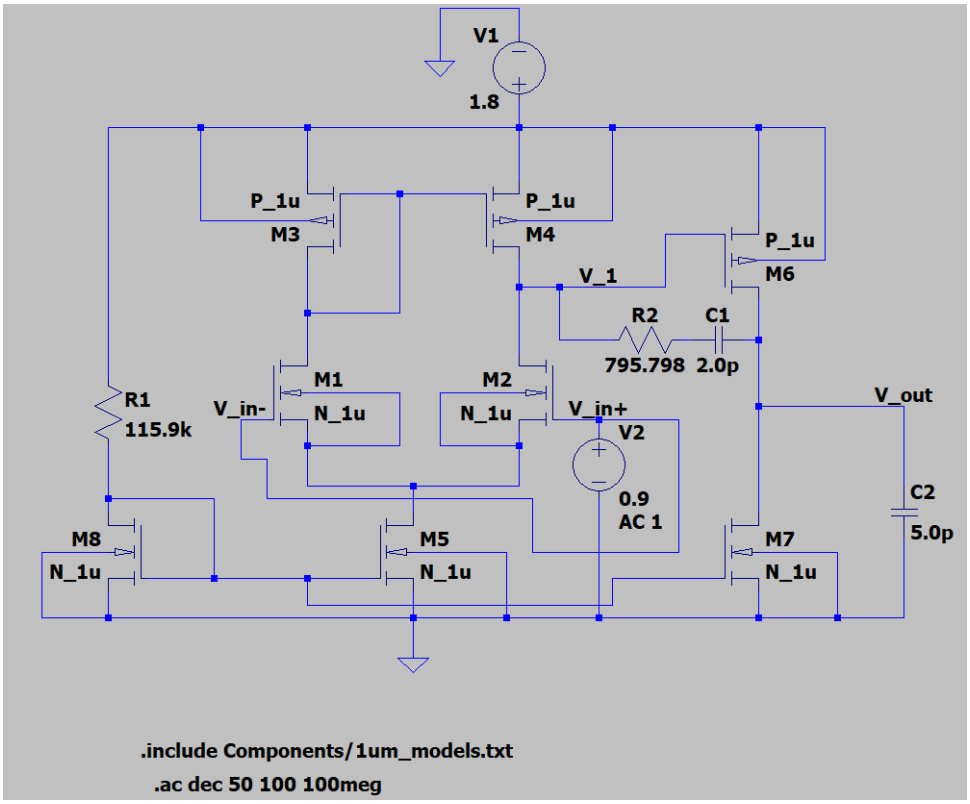


Figure 11: CMRR Analysis Testbench

The differential-mode gain at 100 Hz is shown in Fig. 6. The common-mode gain was measured at the same frequency. The measured value of −8.854 dB is shown in Fig. 12.

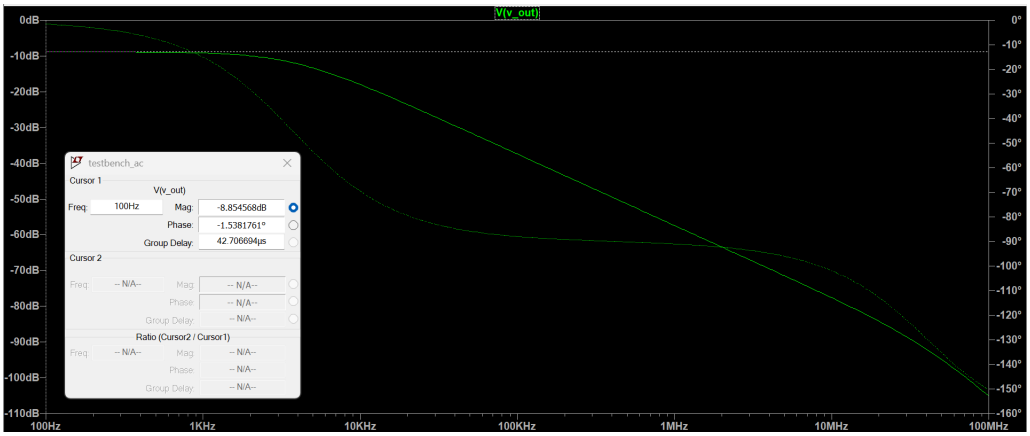


Figure 12: Common Mode Differential Gain Measurement at 100 Hz

Using the simulated differential-mode gain of 68.526 dB and common mode differential gain of −8.854 dB, the CMRR is calculated as

$$\begin{aligned} \text{CMRR}_{\text{dB}} &= A_{d,\text{dB}} - A_{cmdm,\text{dB}} \\ &= 68.526 - (-8.854) \\ &= 77.38\,\text{dB} \end{aligned}$$

The simulated CMRR is therefore 77.38 dB, indicating that the OTA provides substantial rejection of common-mode signals.

Power Supply Rejection Ratio (PSRR) Analysis

The Power Supply Rejection Ratio (PSRR) measures the ability of an OTA to reject variations in its supply voltage and prevent them from affecting the output. A higher PSRR indicates better rejection of supply noise and voltage fluctuations.

The PSRR is defined as the ratio of the differential-mode gain (A_d) to the gain from supply voltage variations to the output (A_{vdd}). For the positive supply, it is expressed as

$$\text{PSRR}_{+} = \left| \frac{A_d}{A_{vdd}} \right|$$

Expressing this ratio in decibels gives

$$\begin{aligned} \text{PSRR}_{+,\text{dB}} &= 20\log_{10} \left| \frac{A_d}{A_{vdd}} \right| \\ &= 20\log_{10} |A_d| - 20\log_{10} |A_{vdd}| \\ &= A_{d,\text{dB}} - A_{vdd,\text{dB}} \end{aligned}$$

So, $PSRR_{dB}$ can be calculated using $A_{d,dB}$ and $A_{vdd,dB}$ provided both gains are measured at the same frequency.

The PSRR was evaluated using `testbench_psrr.asc`. The testbench schematic used for this analysis is shown in Fig. 13.

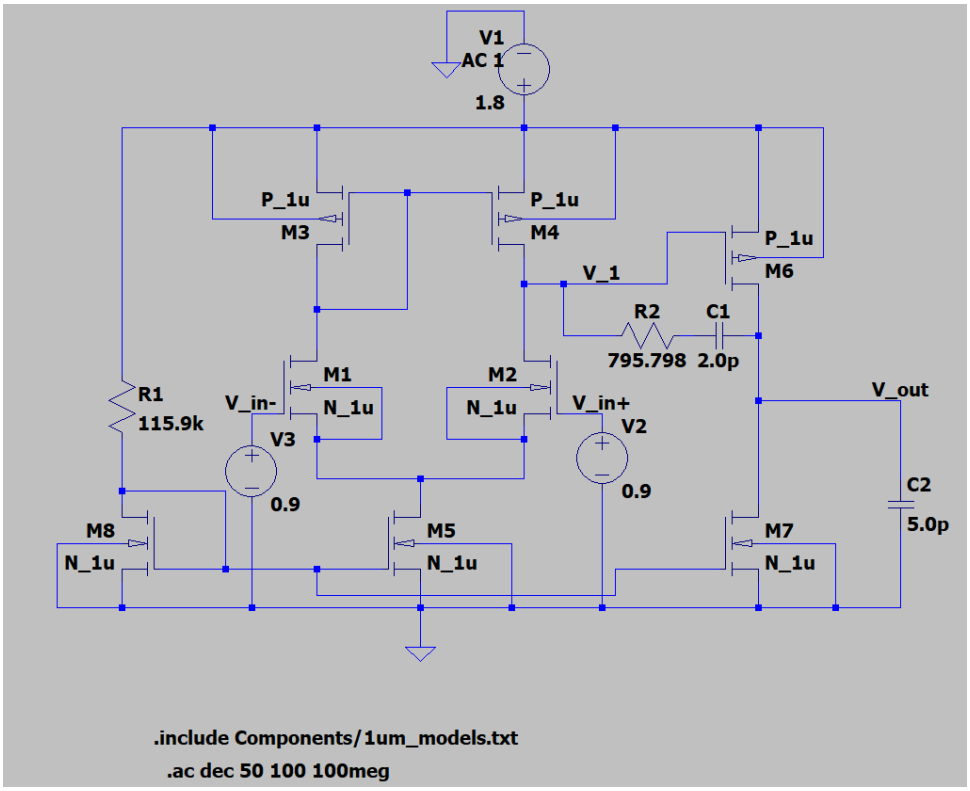


Figure 13: PSRR Analysis Testbench

The simulated supply rejection response is shown in Fig. 14. The measured gain at 100 Hz was −4.544 dB.

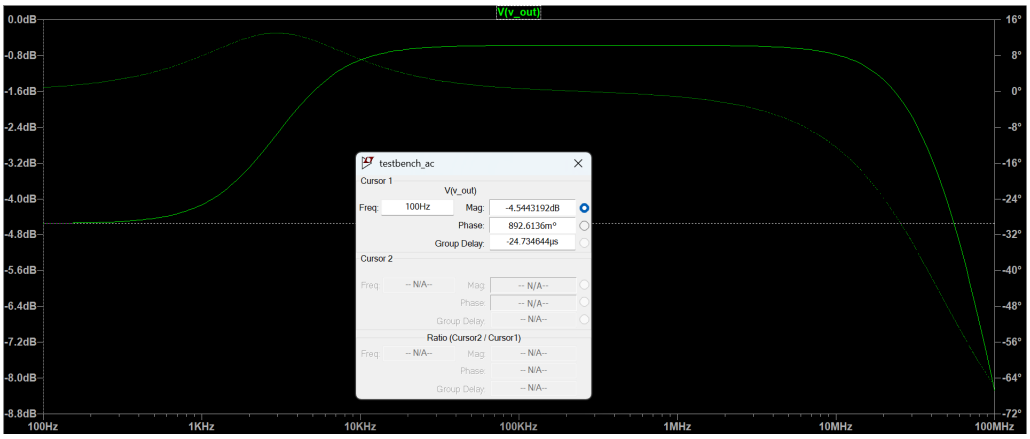


Figure 14: Supply-to-Output Gain Measurement at 100 Hz

Using the simulated differential-mode gain of 68.526 dB and supply-to-output gain of −4.544 dB, the PSRR is calculated as

$$\begin{aligned} \mathrm{PSRR}_{+,\mathrm{dB}} &= A_{d,\mathrm{dB}} - A_{vdd,\mathrm{dB}} \\ &= 68.526 - (-4.544) \\ &= 73.07\,\mathrm{dB} \end{aligned}$$

The simulated PSRR is therefore 73.07 dB, indicating that the OTA provides substantial rejection of variations in its positive supply voltage.

Overall Performance Summary

The performance of the designed two-stage CMOS OTA was evaluated through DC operating-point, AC, transient, ICMR, CMRR, and PSRR analysis. The simulated results and their implications are summarized in Table 9.

Parameter	**Target**	**Simulated**	**Remarks**
DC gain	$\geq 60\,\text{dB}$	$68.525\,\text{dB}$	Meets target.
GBW	$\geq 10\,\text{MHz}$	$9.719\,\text{MHz}$	2.81% below the target.
Phase margin	$\geq 60^\circ$	78.4°	Meets target; indicates good phase stability.
Slew rate	$\geq 5\,\text{V}/\mu\text{s}$	$4.959\,\text{V}/\mu\text{s}$	0.82% below the target.
Settling time (0.1%)	$\leq 500\,\text{ns}$	$100.53\,\text{ns}$	Meets target with settling time well below the limit.
ICMR	0.8–1.6 V	0.410–1.658 V	Covers the specified input range.
Power Consumption	$\leq 500\,\mu\text{W}$	$227.86\,\mu\text{W}$	Meets target.
CMRR	–	$77.38\,\text{dB}$	Provides common-mode signal rejection.
PSRR	–	$73.07\,\text{dB}$	Provides rejection of V_{DD} variations.
Output voltage swing	–	0.410–1.645 V	Measured output range under simulated conditions.

Table 9: Overall Simulated Performance

Overall, the OTA meets most of its specified performance targets. Further optimization is primarily required to improve the GBW and slew rate.

An Interesting Design Trade-off: Lead Compensation

In the original configuration, R_z has been selected in such a way that the RHP zero lies at ∞. But R_Z in can also be adjusted to introduce a left-half-plane (LHP) zero, providing a phase-lead effect that improves the stability of the OTA.

When $R_Z > 1/g_{m6}$, the zero is located at

$$\omega_z = \frac{1}{C_C\left(R_Z - \frac{1}{g_{m6}}\right)}$$

This zero is in LHP and introduces positive phase shift, which can counteract the phase lag associated with the second non-dominant pole. Overlapping ω_{p2} with this zero allows the system to achieve higher GBW and PM.

The second non-dominant pole is approximated by

$$\omega_{p2} \approx \frac{g_{m6}}{C_L}$$

To cancel the effect of ω_{p2}, we set

$$\omega_z = \omega_{p2}$$
$$\frac{1}{C_C\left(R_Z - \frac{1}{g_{m6}}\right)} = \frac{g_{m6}}{C_L}$$

Rearranging gives R_Z as:

$$R_Z = \frac{1}{g_{m6}}\left(1 + \frac{C_L}{C_C}\right)$$

Substituting $C_L = 5\,\text{pF}$, $C_C = 2\,\text{pF}$, and $g_{m6} = 1256.6\,\mu\text{S}$ gives

$$\begin{aligned} R_Z &= \frac{1}{1256.6 \times 10^{-6}}\left(1 + \frac{5}{2}\right) \\ &\approx 795.798 \times 3.5 \\ &\approx 2.785\,\text{k}\Omega \end{aligned}$$

The AC response obtained with the modified compensation resistor is shown in Fig. 15.

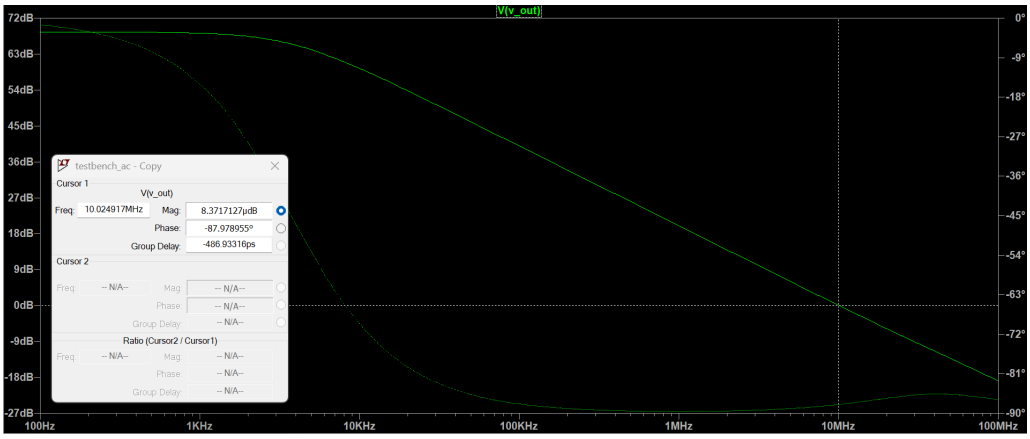


Figure 15: AC Response After Lead Compensation

The unity-gain crossover frequency was measured as 10.025 MHz, with a phase of $-87.979°$. The resulting phase margin is

$$\text{PM} = 180° - 87.979° \approx 92.021°$$

Although the phase margin and bandwidth improved, the transient response showed a reduction in slew rate, as shown in Fig. 16.

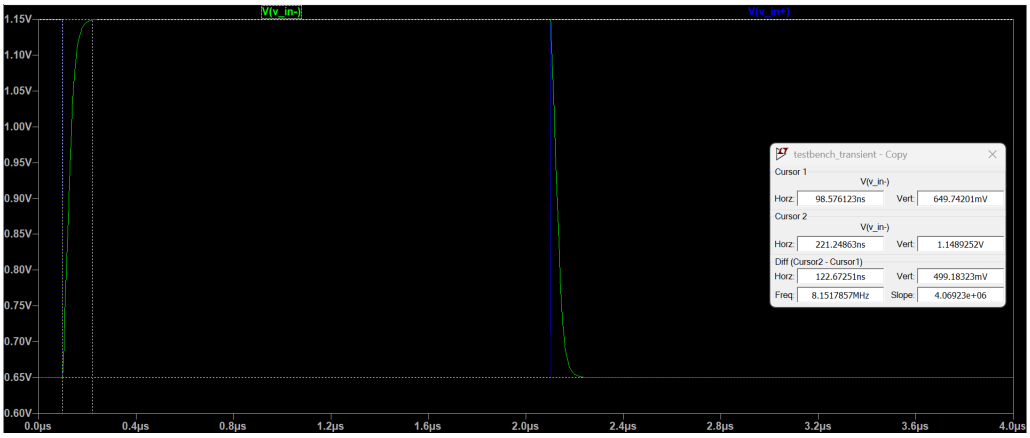


Figure 16: Transient Response After Lead Compensation

Using the two cursor measurements from the transient response, the slew rate was calculated as $4.069\,\text{V}/\mu\text{s}$. The performance of the original and modified compensation configurations is compared in Table 10.

Parameter	**Original**	**Modified**	**Change**
Nulling resistor, R_Z	$795.8\,\Omega$	$2.785\,\text{k}\Omega$	Increased
GBW	9.719 MHz	10.025 MHz	+3.15%
Phase margin	78.4°	92.021°	+13.621° (+17.37%)
Slew rate	$4.959\,\text{V}/\mu\text{s}$	$4.069\,\text{V}/\mu\text{s}$	$-0.890\,\text{V}/\mu\text{s}$ (−17.95%)

Table 10: Performance Comparison Before and After Lead Compensation

This performance drop occurs because increasing R_Z to $2.785\,\text{k}\Omega$ increases the time constant ($\tau = R_Z C_C$) of the compensation branch. During sudden changes, this increase in τ causes the time required for charging and discharging C_C to increase which results in a drop in slew rate.

This demonstrates that the compensation network parameters must be chosen based on overall system requirements, rather than focusing on boosting certain performance parameters.

Conclusion

A two-stage CMOS Operational Transconductance Amplifier (OTA) was designed and analyzed in LTspice using a 180 nm CMOS process framework, with a 1.8 V supply and a 5 pF load capacitance.

Its performance was evaluated through DC operating-point, AC, transient, ICMR, output swing, CMRR, and PSRR analysis.

The OTA achieved a DC gain of 68.525 dB, a phase margin of 78.4°, a GBW of 9.719 MHz, and a slew rate of 4.959 V/μs.

Increasing the series resistor R_Z to 2.785 kΩ canceled the non-dominant pole ω_{p2}, improving the phase margin to 92.021° and the GBW to 10.025 MHz. However, the larger τ slowed down the charging of C_C, reducing the slew rate to 4.069 V/μs. This demonstrates that rather than focusing on boosting certain performance parameters, compensation network parameters must be chosen based on overall requirements of the system.

References

The following resources were consulted to understand the analog circuit design principles and analytical methods used in this project.

- **Differential Amplifier Design — Hafeez KT.** Lecture playlist. Topics: differential-pair analysis and transistor sizing.
- **Two-Stage Op-Amp Design — Hafeez KT.** Lecture playlist. Topics: Miller compensation, second-stage design, and small-signal modelling.
- **Two-Stage CMOS Op-Amp, Chapter 5 - Anonymous** Read Chapter 5. Used for the additional derivation of the compensation-capacitor selection guideline based on load capacitance.